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higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

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NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

2 August (X-Paper)
09:00 – 12:00

This question paper consists of 7 pages and a 5-page formula sheet.

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Questions may be answered in any order, but subsections of questions must be kept together.
5. Show ALL the intermediate steps. Simplify where possible.
6. ALL graph work must be done in the ANSWER BOOK.
7. Questions must be answered in BLUE or BLACK INK.
8. Write neatly and legibly.

QUESTION 1

1.1 Determine the values of the following limits:

1.1.1 $$\lim_{x \to \frac{\pi}{4}} \left(\frac{1 - \tan^2 x}{\frac{\pi}{8} - \frac{1}{2}x} \right)$$ (4)

1.1.2 $$\lim_{x \to 2} \left(\frac{x-2}{\ln\left(\frac{x}{2}\right)} \right)$$ (2)

1.2 Determine the value(s) of x where $f(x) = \frac{x}{x^2 + 5x + 6}$ is discontinuous. (2)

[8]

QUESTION 2

2.1 Given: $y = \arccos x$

2.1.1 Make a neat sketch of the graph $y = \arccos x$ for the range $[O; \pi]$. (2)

2.1.2 Prove that if $y = \arccos x$, then $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$. (3)

2.2 Calculate $\frac{dy}{dx}$ of the following, by making use of the derivative of $\sin x$ and $\cos x$, as well as, the rules of differentiation:

$y = \tan x$ (3)

2.3 Determine $\frac{dy}{dx}$ in each of the following cases:

(Simplification is NOT required.)

2.3.1 $y = \frac{e^x}{(\ln^2 x)}$ (3)

2.3.2 $y = x^3 \,.arc\cot x$ (3)

2.3.3 $y = \sqrt[4]{x^4 + \sec 7x}$ (3)

PTO

2.4 Determine the slope of the tangent line at the point (1;2) of the graph:
$\frac{1}{y^2} - \frac{1}{x^2} = y + x.$ (4)

2.5 Calculate $\frac{dy}{dx}$ if $y = 4^x \cdot e^{-x}$ with the logarithmic differentiation. (4)

[25]

QUESTION 3

3.1 The sum of the slant height (h_s) and the perpendicular height ($h_\perp$) of a right circular cone is 20 cm. Calculate the radius and the perpendicular height of the cone, if the volume is to be a maximum.

HINT: Volume of a cone $\frac{1}{3}\pi r^2 h_\perp$ and $h_s^2 = h_\perp^2 + r^2$ (6)

3.2 The motion of a point on a spring is modeled according to the equation,

$s(t) = 3e^{-2,5t} \cdot \cos 2\pi t$

where s is measured in centimetres and t in seconds.

Determine the following:

3.2.1 The displacement after TWO seconds (2)

3.3.2 The velocity after t seconds (2)

3.3 Given:

$f(x) = -x^3 - 2x^2 + 3x + 1$

3.3.1 Determine the coordinates of the turning points of $f(x)$. (2)

3.3.2 Draw up a table of values of x and $f(x)$, with x ranging from $x = -4$ to $x = 2$. (2)

3.3.3 Draw a neat graph of $f(x)$ between the values and show the turning points on it. (2)

3.3.4 Use the TABLE and the GRAPH to estimate a value for the root between $x = -1$ and $x = 0$ of the equation $-x^3 - 2x^2 + 3x + 1 = 0$ and use Taylor/Newton method to determine a better approximation of this root (the root must be correct up to THREE decimal figures). (4)

[20]

QUESTION 4

4.1 Determine $\int y.dx$ if:

4.1.1 $y = e^{2x}.\sin x$ (4)

4.1.2 $y = \dfrac{\pi^{\ln x}}{x}$ (3)

4.1.3 $y = \dfrac{1}{25 + 64x^2}$ (3)

4.1.4 $y = \cos 7x.\sin 5x$ (3)

4.1.5 $y = \dfrac{\cos 2x}{\sin x + \cos x}$ (4)

4.2 Determine $\int y.dx$ by resolving the integrals into partial fractions:

$$y = \frac{x+1}{x^2 - 10x + 25}$$ (5)

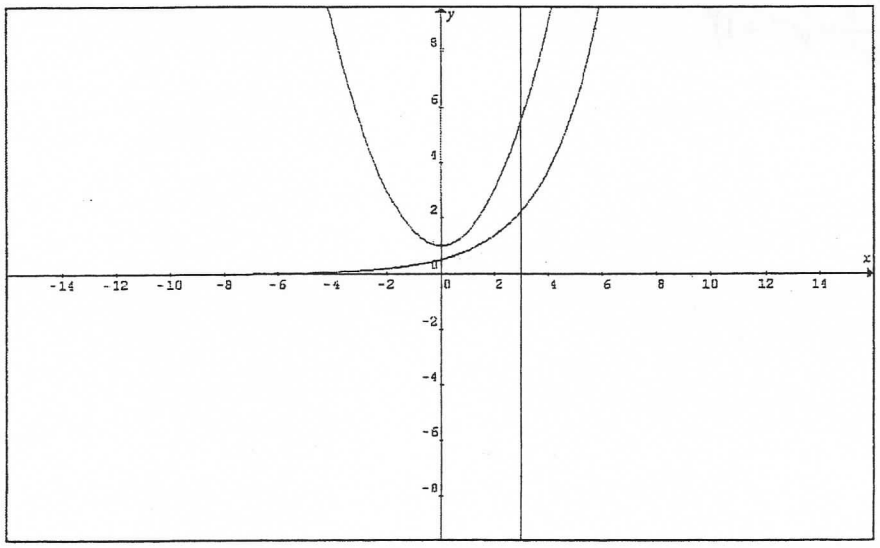
[22]

QUESTION 5

5.1 Determine $\int_0^{\infty} e^{-st}.f(t).dt$ if $f(t)=-\frac{1}{4}$. (4)

5.2 The sketch below shows the area bounded by the curves:

$y=\frac{1}{2}e^{\frac{1}{2}x}$, $y=\frac{1}{2}x^2+1$, $x=3$ and the y-axis.



5.2.1 Draw the sketch in the ANSWER BOOK and show on the sketch the enclosed area, the representative strip, as well as, the lower and upper limits. (3)

5.2.2 Calculate the magnitude of the area in QUESTION 5.2.1. (3)

5.2.3 Calculate the volume generated when the area in QUESTION 5.2.2 rotates about the x-axis. (4)

5.3 Determine the second moment of mass of a rectangular lamina of mass m about an axis parallel to ONE side of the lamina. (4)

[18]

QUESTION 6

6.1 If the gradient of a curve is $\frac{dy}{dx}=\frac{3x+1}{3y}$, find the equation for the curve if it is known that $y=1$ and $x=-1$. (4)

6.2 Determine the general solution of the following:

$$\frac{d^2y}{dx^2}=\left(e^{-x}+1\right)^2$$

(3)
[7]

TOTAL: 100

MATHEMATICS N5

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \operatorname{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = ½ - ½ \cos 2A$

$\cos^2 A = ½ + ½ \cos 2A$

$\sin (A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$\sin A \cos B = ½ [\sin (A + B) + \sin (A - B)]$

$\cos A \sin B = ½ [\sin (A + B) - \sin (A - B)]$

$\cos A \cos B = ½ [\cos (A + B) + \cos (A - B)]$

$\sin A \sin B = ½ [\cos (A - B) - \cos (A + B)]$

$$\tan x = \frac{\sin x}{\cos x} ; \sin x = \frac{1}{\operatorname{cosec} x} ; \cos x = \frac{1}{\sec x}$$

BINOMIAL THEOREM

$$(x+h)^n = x^n + nx^{n-1}\,h + \frac{n(n-1)}{2!}x^{n-2}\,h^2 + \ldots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x)\,.\,v(x)$$

$$\frac{dy}{dx} = u\,.\,\frac{dv}{dx} + v\,.\,\frac{du}{dx}$$

$$= u \cdot v' + v \cdot u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v\,.\,\frac{du}{dx} - u\,.\,\frac{dv}{dx}}{v^2}$$

$$= \frac{v \cdot u' - u \cdot v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x.\ln a$	$\frac{a^x}{\ln a} + c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{x \ln a}$	—
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$
$\tan x$	$\sec^2 x$	$\ln(\sec x) + c$
$\cot x$	$-\text{cosec}^2 x$	$\ln(\sin x) + c$
$\sec x$	$\sec x \cdot \tan x$	$\ln[\sec x + \tan x] + c$
$\text{cosec } x$	$-\text{cosec } x \cdot \cot x$	$\ln(\text{cosec} x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	—
$\text{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2 - x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{a^2 + x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{x\sqrt{x^2 - a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} + c$
$\frac{1}{x^2 - a^2}$	—	$\frac{1}{2a} ln\left(\frac{x - a}{x + a}\right) + c$
$\frac{1}{a^2 - x^2}$	—	$\frac{1}{2a} ln\left(\frac{a + x}{a - x}\right) + c$

INTEGRATION

$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x)\, g(x)\, dx$

$$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$$

$$\int [f(x)]^n f'(x)\, dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

$$\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$$

$$\frac{f(x)}{(x+a)^n} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3} + \cdots \frac{Z}{(x+a)^n}$$

APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b y dx \; ; \; A_x = \int_a^b (y_1 - y_2)\, dx$$

$$A_y = \int_a^b x dy \; ; \; A_y = \int_a^b (x_1 - x_2)\, dy$$

VOLUMES

$$V_x = \pi \int_a^b y^2 dx \; ; \; V_x = \pi \int_a^b \left(y_1^2 - y_2^2\right) dx$$

$$V_y = \pi \int_a^b x^2 dy \; ; \; V_y = \pi \int_a^b \left(x_1^2 - x_2^2\right) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2 dA \; ; \; I_y = \int_a^b r^2 dA$$

MOMENTS OF INERTIA

Mass = density × volume

$$M = \rho V$$

DEFINITION: $I = m r^2$

GENERAL: $I = \int_a^b r^2 dm = \rho \int_a^b r^2 dV$